



Excessive Idling: How to Determine Idle Times and Associated Fuel Costs

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

Excessive Idling: How to Determine Idle Times and Associated Fuel Costs	0
1.0 Introduction	1
2.0 Best Practice Description	2
3.0 Implementation Steps	2
4.0 Benefits	3
5.0 Resources Required	5
6.0 Supporting Attachments / References	5
7.0 Related Best Practices	5
8.0 Acknowledgements	5

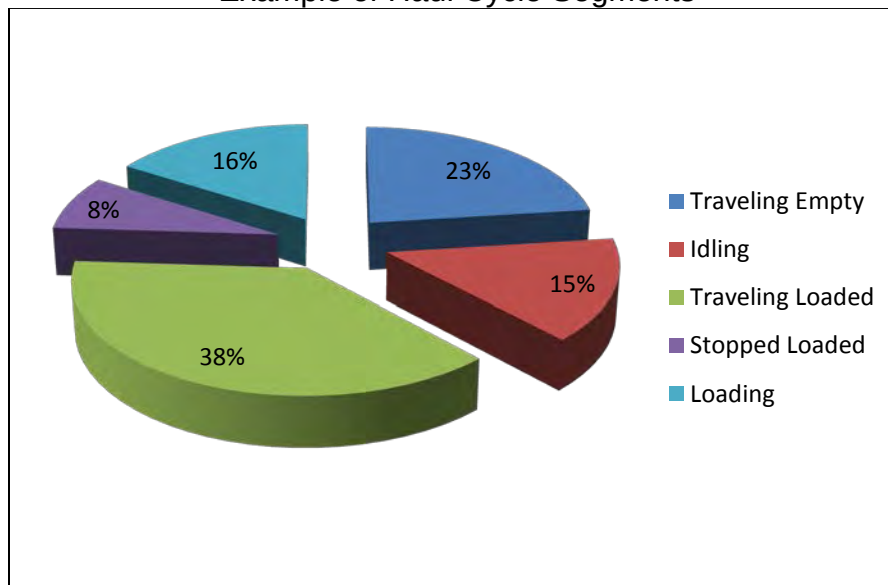
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1.0 Introduction

One of the most important focus areas for mine site operations teams is to maximize haul truck “Traveling Loaded Time” while minimizing inefficient time spent idling and stopped loaded. Not only is this time not spent producing, the unnecessary fuel consumption is costly.

Pit depth, haul distance, ore type, and other variables will govern the percentages that a mining haul truck would spend in each segment of the haul cycle. Flatter or downhill hauls would alter the % time spent traveling loaded or unloaded. Length of haul and price of ore might increase the loading % time as shovels put half loaded buckets on trucks in order to hit target payload on long hauls. With some variation for the above factors, below would be an example of an acceptable breakdown by haul segment type per industry norms.

Example of Haul Cycle Segments



As mines focus more and more on reducing fuel consumption for both cost and environmental purposes, we are forced to optimize our haul cycles and cut out any waste. Excessive idling is an excellent place to start.

Excessive idling is often times merely an operational inefficiency. Trucks are left idling during lunch breaks, mid-shift breaks, refueling, blasting, shift change, and other necessary mine activities. There can also be environmental reasons. Those working in extreme climates are most familiar with excessive idling often times corresponding to seasons. During the winter time, trucks are left idling longer so that engines don’t require a cold restart and cabs stay warm. Conversely, in the warm summer months, trucks stand idling so that air conditioners can be left running. While these problems are real, more practical, cost effective measures can be taken to keep operators comfortable.

Properly identifying, tracking, and putting in place a system to limit machine idling has been elusive on many sites. VIMS data and MineStar Health reporting offers an effective solution for decreasing idling and site fuel consumption.

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Given the above, the importance of following proper shutdown and start-up procedures cannot be stressed enough. Shutting down the machine while the engine is still hot, not allowing it sufficient time to cool can lead catastrophic failure or reduced life. Likewise, cold engine starts without the use of engine preheating can also lead to reduced engine life. Incremental fuel savings should not be substituted for reduced engine life. Proper procedures from machine OMMs and industry best practice should be followed relative to machine shutdown and start-up procedures. There may be good common sense to allowing machines to idle in certain conditions and situations. The objective of this Best Practice is to limit inefficient, lengthy, and unnecessary idling when it occurs.

2.0 Best Practice Description

Using VIMS data and MineStar Health software, the user will be able to trend as much historical data that is available, separating out fleet idling time as a percentage of total operating time. If desired, this can then be broken down further:

- Seasonally
- By Shift
- By Crew
- By Operator

Once fleet idle times have been tracked and any excessive areas (certain crews or operators, for example) have been identified, site can set about eliminating abusive idling. Strategies will vary by site and climate. Solutions to excessive idling could include one or a combination of the following:

- Employing the automatic shutdown timer switch or lowering the time to shutdown setting.
- The mine changing operator Standard Work Procedures (SWPs) to include maximum idling time.
- Utilizing pit supervisors to enforce obligatory engine shutdowns during long truck stop periods.

3.0 Implementation Steps

To implement this download the report (available via the MineStar Reporting KN site).

1. Open the report in Web Intelligence and modify the object _FuelCost to give the correct price of fuel for the site.
2. Ensure that the correct machines are assigned to the correct Fleet. This is done in MineStar Client>Machine Finder
3. Run the report over a period of at least 6 months and save as PDF.

For any questions, please contact MineStar Support

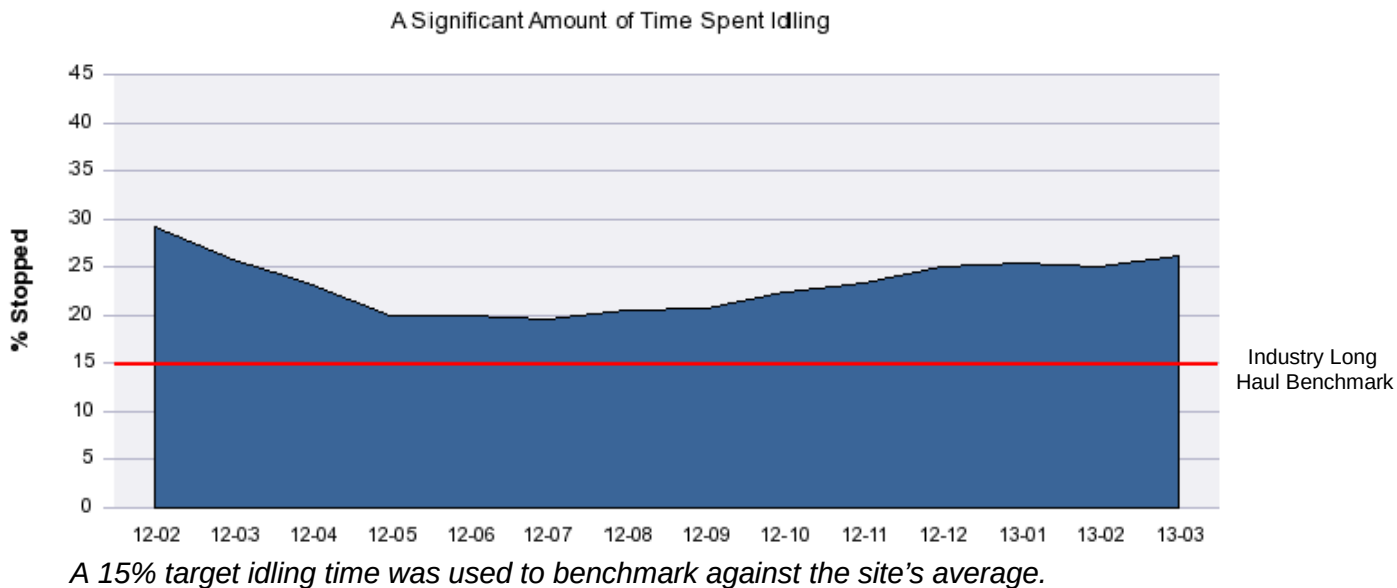
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4.0 Benefits

Whether the reasons are purely operational inefficiency, environmental, or a combination of both, giving the operations department access to this information empowers them to:

- Decrease fuel consumption by controlling machine operators idling times
- Optimize their fleet if the idling is a result of being over-trucked
- Increase production by reallocating trucks that may be underutilized.

As the below example shows, this mine saw increased idling and fuel consumption during the cooler months as operators kept trucks running to avoid cold starts and to keep the cab warm. Idling as a percentage of total operating time was still too high when they reached the summer months. This we were able to trend and benchmark against during the following year.



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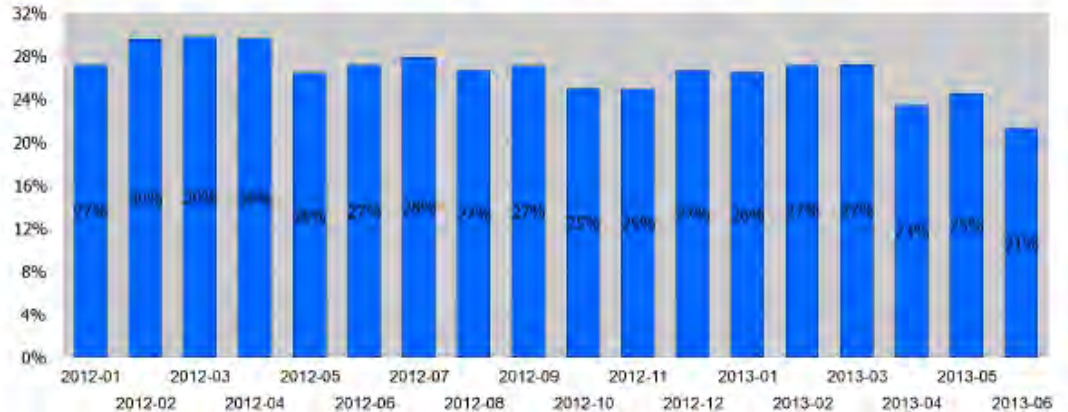
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We had similar findings on another site. Here an evenly split fleet of 50 x793Ds and 785Cs were found to be idling an average of 27% over an 18 month period.

% of Cycle Idling Per Month (Excluding Loading) (Average: 27%)



MineStar report showing average fleet idle time as a percentage of total operating time

Using Health reporting, we can show the customer the cost of idling over an accepted amount. With a target of 12%, this site was spending roughly \$138,000 and \$304,000 on excessive idling for its 785C and 793C fleets, respectively.

Class	Total Cycle Duration	Idle%	Hours Idling	Idle Fuel Burn (L/hr)	Fuel Cost (\$/L)	Total Idle Cost to Site
Cat785C	144585:20:39	25%	35,878.09	10	0.75	\$269,085.69
Cat793D	178614:51:17	28%	50,434.25	14	0.75	\$529,559.6

Class	Idle %			Excessive Idling Hrs.	Idle Fuel Burn (L/hr)	Fuel Cost (\$/L)	Estimated Loss (\$)
	Actual	Target	Delta				
Cat785C	25%	12%	13%	18,527.85	10	0.75	\$138,958.88
Cat793D	28%	12%	16%	29,000.46	14	0.75	\$304,504.88

It is also possible to break down idle time even further. With the use of Minestar Fleet or another fleet management software, we can track idle times by:

- Idle time after machine starts
- Idle time at shovel or loading tool waiting to load
- Stopped and idling during haul to dump (intersection, weigh scales, etc.)
- Idling at the dump
- Idling during break and shift change (no operator in the cab)

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			NUMBER 1308-2.04-1322

Customers and dealers can most effectively take action to limit excessive idle fuel burn firstly by tracking and documenting of % idle time. Areas where the inefficiencies are most apparent (shift change, the dump, operational downtime like blasting, etc.) must be reviewed and a policy implemented to limit idling in these areas. Local conditions, specifically ambient temperatures, must be taken into account. Site must then reexamine idle times once new idling policies have been implemented to gauge improvements.

While a certain level of idling is unavoidable, between 5%-15% depending on site conditions, customer, dealer, and Caterpillar teams can focus on decreasing idling above these industry benchmarks. By showing the % fuel consumption above these norms, and the financial impact of this excessive idling, we can demonstrate the value of tracking and acting on data gathered through VIMS and MineStar.

5.0 Resources Required

- VIMS enabled Caterpillar Off-Highway Trucks
- MineStar Health Software (or in the event that this is not on site, VIMS Supervisor can be used)
- Computer

6.0 Supporting Attachments / References

None

7.0 Related Best Practices

None

8.0 Acknowledgements

Thank you to Joe Atkinson for his assistance with MineStar reporting.

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